Sheet 4 Solution

$\boxed{1}$

a.

$$\begin{array}{r} {}^{1}\;\;5\;1\;2 \\ 7\;2\;3\;+ \\ \hline 1\;4\;3\;5 \end{array}$$

$12 = 1*8+4$

b.

$$\begin{array}{r} {}^{1}\;\;7\;2\;0 \\ 2\;3\;6\;+ \\ \hline 1\;1\;5\;6 \end{array}$$

$9 = 1*8+1$

$\boxed{2}$ a.

$$\begin{array}{r} 7\;\overset{4}{\cancel{5}}\;\overset{9}{1}\;- \\ 2\;2\;4 \\ \hline 5\;2\;5 \end{array}$$

b.

$$\begin{array}{r} \overset{5}{\cancel{6}}\;\overset{11}{\cancel{3}}\;3\;4\;- \\ 7\;3\;2 \\ \hline 5\;4\;0\;2 \end{array}$$

$\boxed{3}$

$$\begin{array}{rrrrrr}
 & 3 & 1 & 2 & & \\
 & 2 & & 1 & & \\
 & & 7 & 2 & 4 & \\
 & & & 4 & 3 & * \\
\hline
 & 2 & 5 & 7 & 4 & \\
3 & 5 & 2 & 0 & & + \\
\hline
3 & 7 & 7 & 7 & 4 &
\end{array}$$

$4 * 3 = 12$

$12/8 = 1$ R 4

$(2 * 3) + 1 = 7$

$7/8 = 0$ R 7

$7 * 3 = 21$

$21/8 = 2$ R 5

$4 * 4 = 16$

$16/8 = 2$ R 0

$(4 * 2) + 2 = 10$

$10/8 = 1$ R 2

$(4 * 7) + 1 = 29$

$29/8 = 3$ R 5

1

b.

$$\begin{array}{rrrrrrr}
2 & & & & & & \\
 & & 1 & 1 & 1 & & \\
1 & & & & 1 & & \\
 & & 6 & 3 & 3 & 4 & \\
 & & 1 & 1 & 3 & 2 & * \\
\hline
 & 1 & 4 & 6 & 7 & 0 & \\
2 & 3 & 2 & 2 & 4 & & + \\
\hline
2 & 4 & 7 & 1 & 3 & 0 &
\end{array}$$

$2 * 4 = 8$

$8/8 = 1$ R 0

$(2 * 3) + 1 = 7/8 = 0$ R 7

$2 * 3 = 6$

$6/8 = 0$ R 6

$2 * 6 = 12$

$12/8 = 1$ R 4

$3 * 4 = 12$

$12/8 = 1$ R 4

$(3 * 3) + 1 = 10 \longrightarrow 10/8 = 1$ R 2

$(3 * 3) + 1 = 10 \longrightarrow 1$ R 2

$(3 * 6) + 1 = 2$ R 3

$\boxed{4}$

a.

$$\begin{array}{ccccl} 1 & 1 & & 1 & \\ 1 & 5 & F & A & \\ & D & A & C & + \\ \hline 2 & 3 & A & 6 & \end{array}$$

$22 = 16 * 1 + 6$

$26 = 1 * 16 + 10$

$19 = 1 * 16 + 3$

b.

$$\begin{array}{ccccl} 1 & & 1 & & \\ 4 & E & 2 & C & \\ 1 & 3 & 9 & 5 & + \\ \hline 6 & 1 & C & 1 & \end{array}$$

$17 = 1 * 16 + 1$

$\boxed{5}$

a.

$$\begin{array}{ccccl} F & 9 & \overset{4}{\cancel{5}} & \overset{18}{2} & \\ & 5 & 2 & 3 & - \\ \hline F & 4 & 2 & F & \end{array}$$

b.

$$\begin{array}{ccccl} 9 & \overset{4}{\cancel{5}} & \overset{17}{\overset{\cancel{1}}{\cancel{2}}} & \overset{19}{3} & - \\ & 2 & 4 & B & \\ \hline 9 & 2 & D & 8 & \end{array}$$

$\boxed{6}$

a.

$$\begin{array}{rrrrr} 5 & 3 & 4 & 4 & \\ & F & 8 & C & B \\ & & & & 6 \ * \\ \hline 5 & D & 4 & C & 2 \end{array}$$

6*11=66
66/16= 4 R 2
(6*12)+4 = 76
76/16 = 4 R12
(6*8)+4= 52
52/16= 3 R 4
(6*15)+3= 93
93/16= 5 R 13

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b.

$$\begin{array}{rrrrr} & & 1 & & \\ & 1 & 4 & & \\ & 1 & 2 & 3 & D \\ 1 & & & 2 & 5 \ * \\ \hline & 5 & B & 3 & 1 \\ 2 & 4 & 7 & A & + \\ \hline 2\ A & 2 & D & 1 & \end{array}$$

13*5=65
65/16= 4 R 1
(5*3)+4= 19
19/16=1 R 3
(5*2)+1 = 0 R 11
5*1=5
5/16= 0 R5

2*13=26
26/16= 1 R 10
(2*3)+1= 7
7/16 = 0 R 7
2*2=4
4/16= 0 R 4
2*1=2
2/16= 0 R 2